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Acicular berthierine-chamosite in coal and lung cancer tissues of Xuan Wei, China

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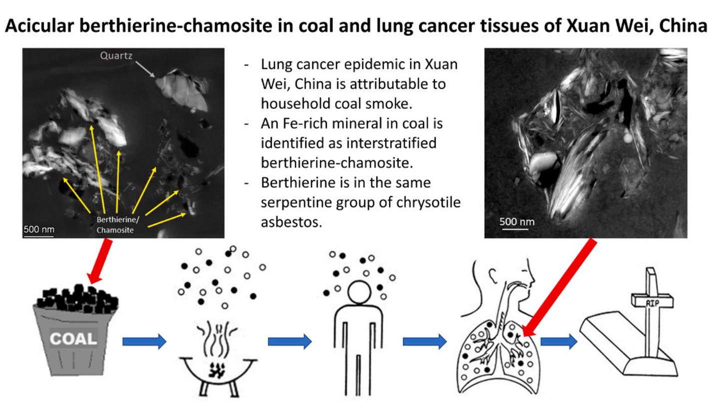
HIGHLIGHTS

- Iron mineral in coal from Xuan Wei lung cancer epidemic area is identified as acicular interlayered berthierine-chamosite.
- Berthierine shares the serpentine group classification with chrysotile (white asbestos).
- Iron leached out of berthierine-chamosite is stored as holo-ferritin in autolysosomes.

GRAPHICAL ABSTRACT



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ABSTRACT

Xuan Wei has China's highest female lung cancer mortality, attributed to household coal smoke exposure. While coal-derived polycyclic aromatic hydrocarbons (PAHs) were previously blamed, persistently rising lung cancer rates despite PAH reductions implicate other carcinogens. Here, we aim to investigate iron-rich minerals in Late Permian coal as potential overlooked drivers. Coal was sampled from Laibin, epicentre of Xuan Wei lung cancer epidemic. Mineral phases in low temperature ash (LTA) were characterized by X-ray diffraction (XRD) along with selected area electron diffraction (SAED) and high-resolution transmission electron microscopy (TEM). Mineral morphology and elemental composition were examined by scanning electron microscope (SEM) and TEM equipped with energy-dispersive X-ray spectrometer (EDS). Lung tissues from 10 patients (5 female, 5 male) were examined for mineral deposition by SEM and TEM. We identified acicular iron-rich interstratified berthierine-

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chamosite minerals coating quartz/calcite in coal. Berthierine shares the serpentine group classification with chrysotile (i.e., white asbestos, a known human carcinogen). We observed abundant berthierine-chamosite needles in cancerous and para-cancerous lung tissues, with iron leaching from the berthierine-chamosite into acidic digestive vacuoles and accumulating in holo-ferritin and hemosiderin aggregates within autolysosomes. Given its much higher iron content than chrysotile, this needle-like berthierine-chamosite in Late Permian coal could be more bioactive and toxic, potentially driving the lung cancer epidemic in Xuan Wei, China. Our findings call for closer collaboration between mineralogists and health scientists for future research to gain deeper mechanistic insights and formulate intervention strategies.

1. Introduction

The carcinogenicity of household coal smoke was first noted in the Chinese county of Xuan Wei, which has the highest women lung cancer mortality in China (Fig. 1) [1]. The findings in Xuan Wei contributed to the evidence base for household coal smoke to be "carcinogenic to humans (Group 1)" [2]. Xuan Wei constitutes a unique natural experiment in environmental carcinogenesis—because its residents used coal mined within a few kilometers of their homes, creating pronounced geographic variations in indoor air pollution exposures along with lung cancer rate disparities among different villages and towns [3]. Excluding smoking and occupational causes, inhalation of bituminous coal (aka "smoky coal") emission was tied to elevated lung cancer risks in earlier epidemiological studies [1]. Inhalation of bituminous coal smoke also increased the incidence of malignant lung tumors in both male and female Kunming mice and Wistar rats [4].

Polycyclic aromatic hydrocarbons (PAHs) in coal smoke has been suspected to be the major group of carcinogens contributing to the exceptionally high lung cancer mortalities in Xuan Wei, China. High concentrations of mutagenic PAHs were found in the emission particles collected during bituminous coal combustion, which is associated with the highest lung cancer risk compared to burning of other fuel types such as wood and anthracite (aka "smokeless coal" locally) [1]; the KRAS and TP53 mutation spectrum in lung tumors, mostly G:C to T:A transversions, were consistent with an exposure to PAHs [5]; lung cancer risks in central Xuan Wei decreased by around 50 % after stove improvement which vented the coal smoke outdoors [6]. More recent studies, however, yielded contrasting evidence that PAHs could not be held responsible for the lung cancer epidemic in Xuan Wei [7,8]. An expanded analysis of 15 types of coal in an ecological correlation study failed to observe any association between PAHs emissions and lung cancer risks [3]; instead of benzo[*a*]pyrene-diol-epoxide (BPDE) which is a PAH metabolite, the oxidatively damaged base 8-Hydroxyguanine (8OHG) was found to be the major cause for G:C to T:A transversions in DNA of human cells [9]; despite large concentration reductions in PAHs, human carcinogen according to IARC, in the residents' homes since 1980s [10], the lung cancer rates in Xuan Wei continued to increase [11,12], in a higher pace than other rural areas of China [12]. Other carcinogens than PAHs could have been overlooked. Can this overlooked carcinogen be iron (Fe)?

Given this natural experiment of wide geographic variations in indoor air pollution exposures and lung cancer rates in Xuan Wei, earlier studies mainly compared domestic bituminous coal use with other major fuels: wood and anthracite. Even within the same category of bituminous coal, however, subtypes of bituminous coal were associated with large variation in lung cancer risks within Xuan Wei [13]. Could such variation in lung cancer risk be explained by the iron content in bituminous coal from the Late Permian era of 255 million years ago? We observed excellent correlation of the iron content in six types of bituminous coal with the logit of lung cancer mortalities (R^2=0.996;

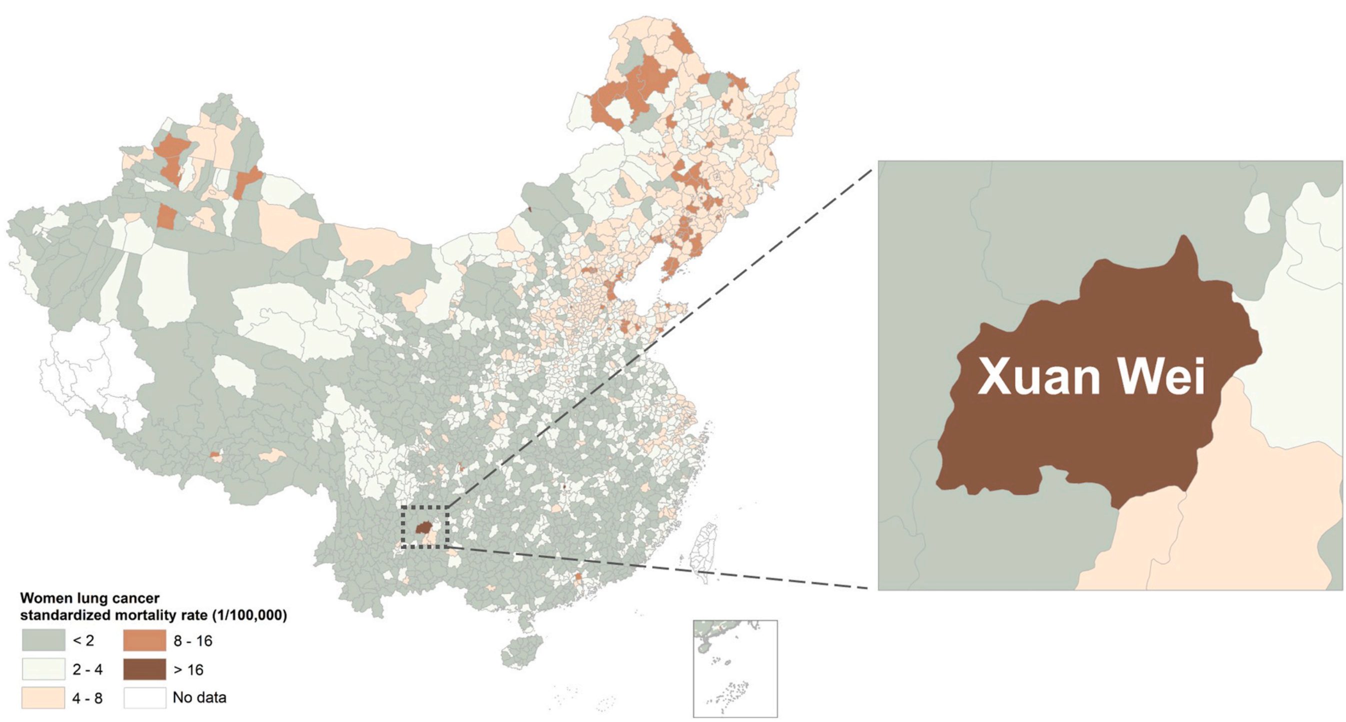

Fig. 1. Map of China showing lung cancer mortality rates of females, in 1973–1975, and the location of Xuan Wei [52]. The mortality rate of female lung cancer in Xuan Wei was higher than those in all other counties and cities in China, being 8 times that of the national average level. Male lung cancer risks in Xuan Wei are similar to those in females and 4 times higher than the national average level.

$p < 0.0001$) across towns of Xuan Wei [14]. Complementing with the human population evidence, our animal experiments demonstrated the effect of deferiprone (DFP, an iron chelator) in reducing the number of neutrophils in the lung lavage of FVB/N mice acutely exposed to Late Permian coal ash samples heated at various temperatures to mimic coal combustion [7]. What kind of iron minerals are in these Late Permian coals?

In the present paper, we report the identification and characterization of acicular berthierine-chamosite (a type of Fe^{2+}-rich silicate) in the Late Permian coal, coal ash, and lung cancer tissue samples from Xuan Wei, China. Findings here deliver novel insights into the coal-derived berthierine-chamosite as potential carcinogen underlying lung cancer epidemic in the study region. Built on the current research work, we will endeavour to further confirm the carcinogenicity of those berthierine-chamosite minerals in bioassays, to test whether iron chelators can be effective in lung cancer risk reduction in rodents and then in a population cohort, and finally to recommend iron chelation and fuel change intervention policies in afflicted areas.

2. Material and methods

2.1. Preparation and mineralogical analysis of coal samples

Coal samples were taken from the working faces of mined coal seams from the town of Laibin (LB) which is the epicenter of the lung cancer epidemic in Xuan Wei, China. In the study area, there are over 30 coal seams from the Late Permian period, and local residents have primarily mined and utilized about 10 of these seams over the past few decades [15–17]. Compared with other Late Permian bituminous coals in Xuan Wei, which all contain the minerals of quartz, calcite, and berthierine-chamosite, the seam of C1 coal has the largest coal ash fraction and the highest iron content [3]. The iron content of C1 coal in Laibin is as high as 3.9 % quantified as Fe_2O_3 [14]. Coal ash from C1 coal was prepared by low temperature ashing (LTA) which involves heating coal samples in oxygen plasma to a temperature below 200°C to remove carbonaceous materials while preventing damage to the mineral crystals of study interest.

Mineral phases in coal and coal ash were measured by X-ray diffraction (XRD, Rigaku SmartLab SE) along with the selected area electron diffraction (SAED) patterns and high-resolution transmission electron microscopy (TEM) images. The XRD analysis was performed on a powder diffractometer with Ni-filtered Cu-Kα radiation and a scintillation detector. The XRD pattern was recorded with a 3–60° 2θ interval and a 0.02° step size. The diffractograms were then subjected to quantitative mineralogical analysis using the commercial interpretation software MDI-JADE Version 6.0. Mineral morphology and elemental composition were examined by scanning electron microscope (SEM) equipped with an energy-dispersive X-ray spectrometer (EDS) and TEM equipped with an EDS.

2.2. Preparation and ultrastructural analysis of lung tissue samples

All lung tissue samples were obtained from therapeutic surgical resections of patients with a confirmed histological diagnosis of primary lung cancer (specifically, lung adenocarcinoma) from Yunnan Cancer Hospital. The consecutive series of 10 lung cancer patients (5 female and 5 male) for research were lifelong residents of the high-risk Xuan Wei area, with documented history of household use of locally sourced bituminous coal but without occupational exposures such as mining, welding, or foundry work. One male patient was a clerical worker, and all the other 9 patients were local farmers.

From each patient, we collected cancerous tissue, para-cancerous tissue, peri-bronchial lymph nodes, and distant normal tissue for pathological and mineralogical analyses. The tissues were carefully excised into 1 mm in length and immediately immersed in 2.5 % glutaraldehyde solution for 4 h at 4°C for immersion fixation. The fixed samples were rinsed three times in 0.1 M cacodylate buffer for 10 min each time and post-fixed in 1 % osmium tetroxide (OsO_4) in 0.1 M cacodylate buffer for 1 h to enhance membrane contrast by staining lipids and stabilizing the ultrastructure. The fixed tissues were dehydrated through a graded series of ethanol solutions (30 %, 50 %, 70 %, 90 %, 95 %, and 100 %, respectively) for 10 min at each concentration. Dehydrated lung tissues were stained with 2 % aqueous uranyl acetate for 15 min at room temperature to enhance contrast by binding to nucleic acids and proteins. Stained tissues were infiltrated with a mixture of propylene oxide and epoxy resin in a 1:1 ratio for 2 h under room temperature, embedded in resin that was polymerized in an oven at 60°C for over 48 h to solidify for sectioning. The embedded lung tissue blocks were trimmed, and 90 nm ultrathin sections were cut using an ultramicrotome (Leica EM UC6 Ultramicrotome) equipped with a diamond knife. The sections were carefully collected onto 200 mesh copper grids coated with Formvar supporting film. The ultrathin sections were stained with Reynolds' lead citrate for 10 min to enhance contrast. The stained sections were examined using TEM (Hitachi HT7800) operated at an accelerating voltage of 120 kV. High-resolution images were captured to analyze the ultrastructural details of lung tissue, including cellular organelles, mineral particle depositions, and any pathological features. This study was approved by the Institutional Review Board of Yunnan Cancer Hospital (KYCS2021113) and a proof/certificate of approval is available upon request.

3. Results

3.1. Identification of berthierine-chamosite in C1 coal

XRD pattern (Fig. 2) from the LTA sample of Late Permian coal identified quartz, calcite, chamosite, and interstratification of berthierine with chamosite. The 7 Å peak in the XRD pattern is sharp and narrow whereas the 14 Å peak is diffuse and broad, which is characteristic of randomly interstratified 14 Å (chamosite) and 7 Å (berthierine) layers [18–20]. According to Moore and Reynolds [18], berthierine is a type of mineral challenging to be identified and characterized: berthierine with a small admixture of chamosite may easily be mistaken as pure chamosite. For better characterization, we used Focused Ion Beam (FIB) to prepare thin foil specimens from the cross-sections of berthierine-chamosite mineral in coal for further examination under TEM. High-resolution TEM images at the top row of Fig. 2 show regions of just 7 Å layer of berthierine, and the image at the bottom row show regions with interstratification of 7 Å layer of berthierine with 14 Å layer of chamosite. Therefore, it is safe to identify this mineral as inter-layered berthierine-chamosite while pure berthierine is occasionally spotted under TEM.

3.2. Morphology and composition of berthierine-chamosite in coal and lung tissue

As seen in Fig. 3, the bulky grains in coal are mostly quartz, while the much finer acicular minerals are berthierine-chamosite as identified by SEM-EDS. Berthierine-chamosite occurs mostly as cell-infillings along with quartz or fills in the cavities of quartz and calcite. Some quartz grains are coated with thin layers of berthierine-chamosite, which may lead to misclassification if we rely on thick specimens or just surface examination. According to SEM-EDS results, the average chemical formula of the berthierine-chamosite in C1 coal was calculated as $(Fe^{2+}_{3.98}Mg_{0.68}Al_{1.35})(Si_{2.56}Al_{1.35})O_{10}(OH)_8$, which is consistent with the general formula of $(Fe^{2+}_3Mg_{0.75}Al_2)(Si_{2.5}Al_{1.5})O_{10}(OH)_8$ for berthierine [18]. Berthierine (serpentine group) and chamosite (chlorite group) have similar chemical composition of Si, Al, and Fe elements [18].

In the resected lung tissue from lung cancer patients, mineral deposition is severe while the deposits are not severe enough for pneumoconiosis diagnosis. In some autolysosomes, berthierine-chamosite has retained its acicular morphology (Fig. 4) while its

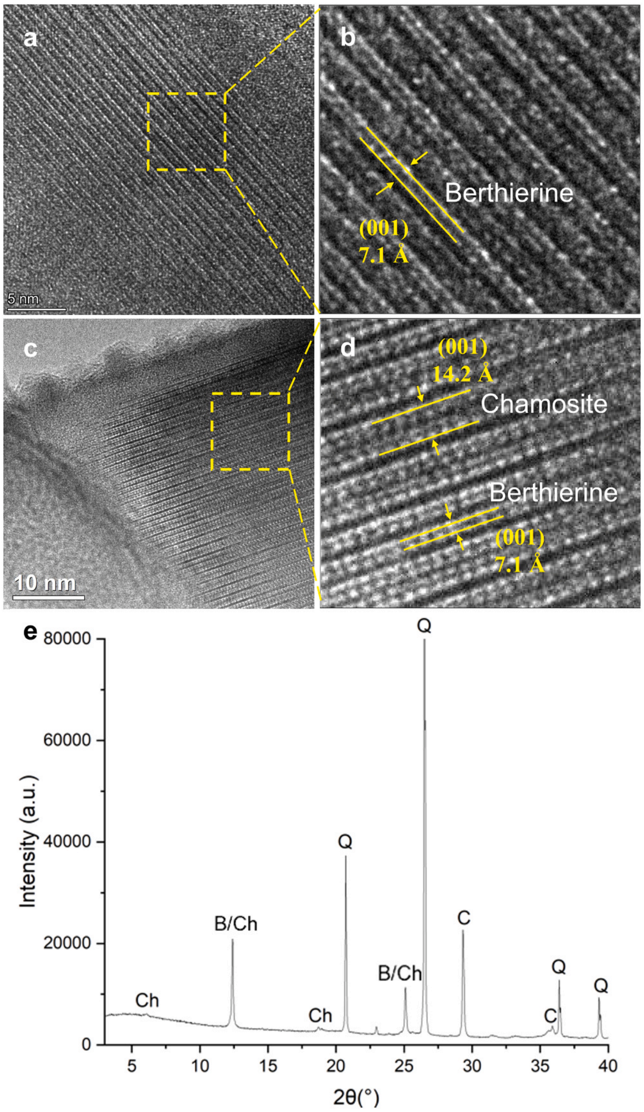

Fig. 2. Identification of inter-layered berthierine-chamosite in Late Permian coal of Xuan Wei, China by high-resolution transmission electron microscopy (TEM) and X-ray diffraction (XRD). (a, b) TEM images show regions of just 7 Å layer of berthierine in coal. (c, d) TEM images show regions with interstratification of 7 Å layer of berthierine with 14 Å layer of chamosite in coal. (e) XRD spectrum of quartz (Q), calcite (C), chamosite (Ch), and interstratification of berthierine with chamosite (B/Ch) in coal; the 7 Å peak is sharp and narrow whereas the 14 Å peak is diffuse and broad, which is characteristic of randomly interstratified 14 Å (chamosite) and 7 Å (berthierine) layers.

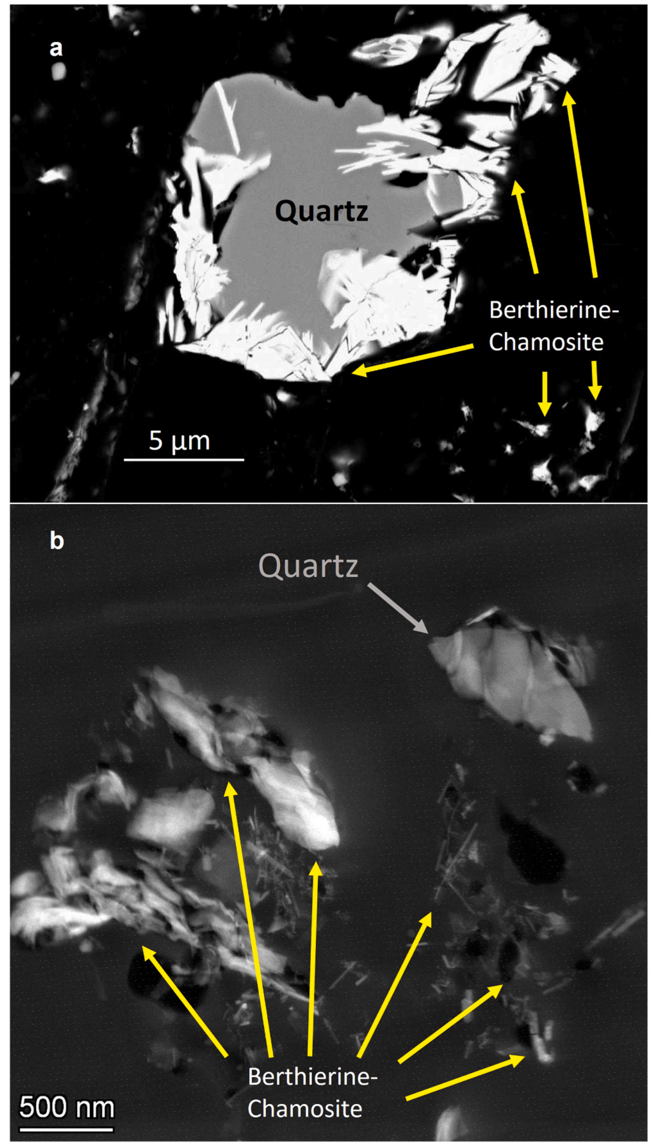

Fig. 3. Morphology of berthierine-chamosite in Late Permian coal of Xuan Wei, China. (a) Backscattered electron scanning electron micrograph (BSE-SEM) observation of the two major minerals, quartz and berthierine-chamosite. (b) Transmission electron microscopy (TEM) observation of quartz and berthierine-chamosite. Some quartz grains are coated with thin layers of berthierine-chamosite, which may lead to misclassification if we rely on thick specimens or just surface examination. While quartz is bulky, berthierine-chamosite exhibits acicular morphologies, occurring as bundled fibrous aggregates or discrete individual fibers.

elemental composition has changed, with preferential depletion of Fe from the mineral. Bundles of needle-shaped berthierine-chamosite are abundant in para-cancerous tissue and distant normal tissue; in the autolysosomes, some needles have become separate and wavy.

3.3. Abundant Fe-rich ferritin in autolysosomes in lung cancer tissues

Especially in lung cancer cells, we have observed abundant granules of Fe-rich ferritin in the autolysosomes under TEM (Fig. 5). Fe leached from the berthierine-chamosite fibers has probably been stored in the Fe-rich ferritin particles in autolysosomes, which appear similar to the "iron-rich inclusion bodies" detected in macrophages after asbestos fiber exposure in mice [21]. Free ferritin molecules in the cytosol that would indicate endogenous ferritin synthesis are substantially absent. The autolysosomes full of Fe-rich ferritin aggregates observed here are consistent with the siderosomes reported by Iancu et al. in their iron overload studies [22–25], and with the exosomes for ferritin transport reported by Truman-Rosentsvit et al. [26] and Ito et al. [27]. Occasionally, acicular berthierine-chamosite minerals are still visible in autolysosomes of cancer cells, while in most cases the autolysosomes contain mainly Fe-rich ferritin clusters. Some Fe-rich ferritin particles can be individually resolved, whereas most particles have coalesced; after degradation of the protein coat, some ferritin particles may form an extremely electron-opaque mass, considered to be hemosiderin. In some lung cancer cells adjacent to the rich capillary vasculature (Fig. 6), the nuclei are surrounded by numerous autolysosomes that contain ferritins of variable electron-density ("iron-rich" vs. "iron-poor") and hemosiderin "clumps" with extreme electron-density.

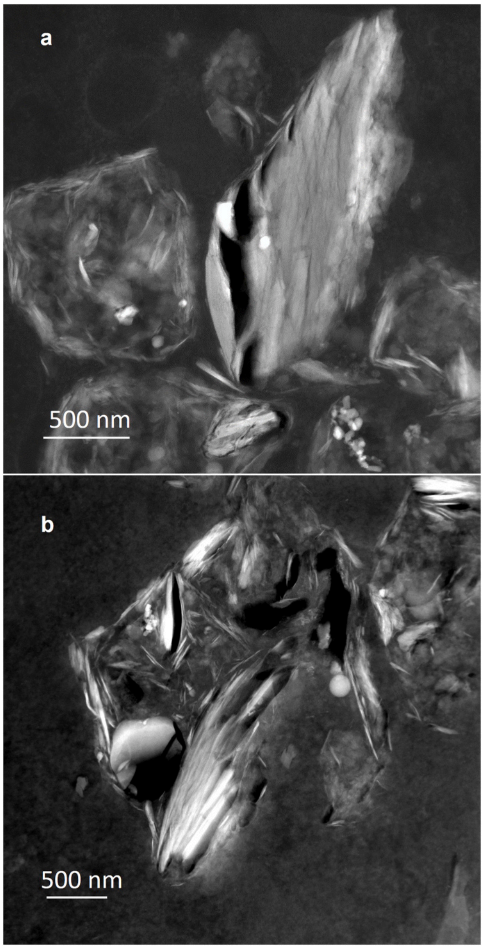

Fig. 4. Abundant deposits of modified berthierine-chamosite in the para-carcinoma tissues of lung cancer cases from Xuan Wei, China. (a, b) Transmission electron microscopy (TEM) images show partially modified berthierine-chamosite fibers in the autolysosomes of para-carcinoma tissues. Although the acicular morphology of berthierine-chamosite retains, its elemental composition has changed, with preferential depletion of iron in the acidic environment of autolysosomes.

4. Discussion

For the first time in the last four decades of search for the culprits underlying the lung cancer epidemic of Xuan Wei, China, we here reported the identification and characterization of acicular berthierine-chamosite in locally consumed Late Permian coal, coal ash, and lung cancer tissues sampled from the epidemic's epicentre. In comparison with the bulky grains of quartz and calcite, the much finer acicular minerals of berthierine-chamosite have a higher chance of entering the deep lungs. Needle-shaped berthierine-chamosite bundles are abundant in para-cancerous and distant normal tissues, and iron-rich ferritin clusters are abundant inside the autolysosomes of lung cancer cells. Our observations of autolysosomes full of Fe-rich ferritin clusters are consistent with the "iron-rich inclusion bodies" in macrophages after asbestos fiber exposure in mice [21], the siderosomes as reported by Iancu et al. in their iron overload studies [22–25], and the exosomes for ferritin transport as reported by Truman-Rosentsvit et al. [26] and Ito et al. [27]. While functional assays are still needed in the future to establish causality between mineral-derived Fe and autolysosomal ferritin storage, our findings tend to suggest a local iron overload in lungs due to inhalation of fine and acicular iron-rich minerals in Xuan Wei, which echoes the hypothesis of Toyokuni [28–30] that carcinogenesis is a process to establish iron addiction with ferroptosis resistance.

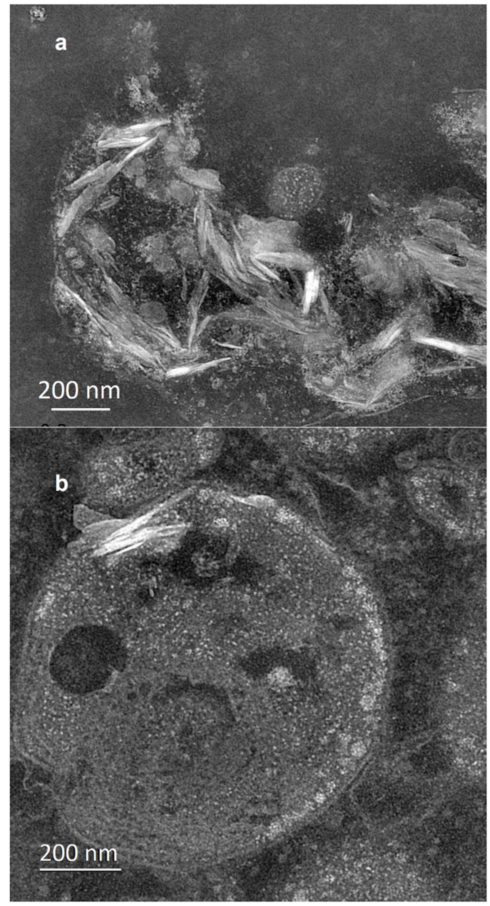

Fig. 5. Inverted images of autolysosomes under transmission electron microscopy (TEM). (a) One autolysosome in which bundles of needle-shaped berthierine-chamosite are abundant along with aggregated or randomly dispersed Fe-rich ferritin particles of high electron density. (b) Another autolysosome in which a few needle-shaped berthierine-chamosite particles are still visible and a large number of Fe-rich ferritin particles align along the phospholipid membranes.

The identification of inter-layered berthierine-chamosite in coal of this lung cancer epidemic area warrants attention because berthierine is in the same serpentine group as chrysotile asbestos. With a chemical formula of $(Fe^{2+}_3Mg_{0.75}Al_2)(Si_{2.5}Al_{1.5})O_{10}(OH)_8$, berthierine (Fe-Al serpentine), is an iron-rich, aluminous, 1:1-type layer silicate of the serpentine group, the same mineral group that chrysotile (white asbestos, the principal varieties of asbestos) belongs to [18]. Chrysotile is classified as a human carcinogen, although there is still controversy in banning chrysotile globally [31]. Divalent iron (Fe^{2+}), an electron donor, on the asbestos fibers causes the formation of oxygen-activated radical species which are extremely aggressive and may play a key role in carcinogenesis [32,33]. Compared to amphibole (the other group of asbestos known as potent human carcinogens), the serpentine chrysotile usually has lower iron content [34], but Fe^{2+} ions are particularly common as a substitute for magnesium in Canadian chrysotile which demonstrates high carcinogenicity in animal experiments [35]. Moreover, chrysotile is associated with a shorter dissolution time than amphiboles and hence a comparable release of iron; in terms of surface availability of iron, the overall toxicity potential of chrysotile is not lower than that of amphibole asbestos [36].

In the same serpentine group, berthierine can be more active than chrysotile. Different from chrysotile, berthierine has an octahedral sheet containing essentially Fe^{2+} and not Mg^{2+}. The excess incidence of lung cancer among iron miners was associated with the free radicals arising

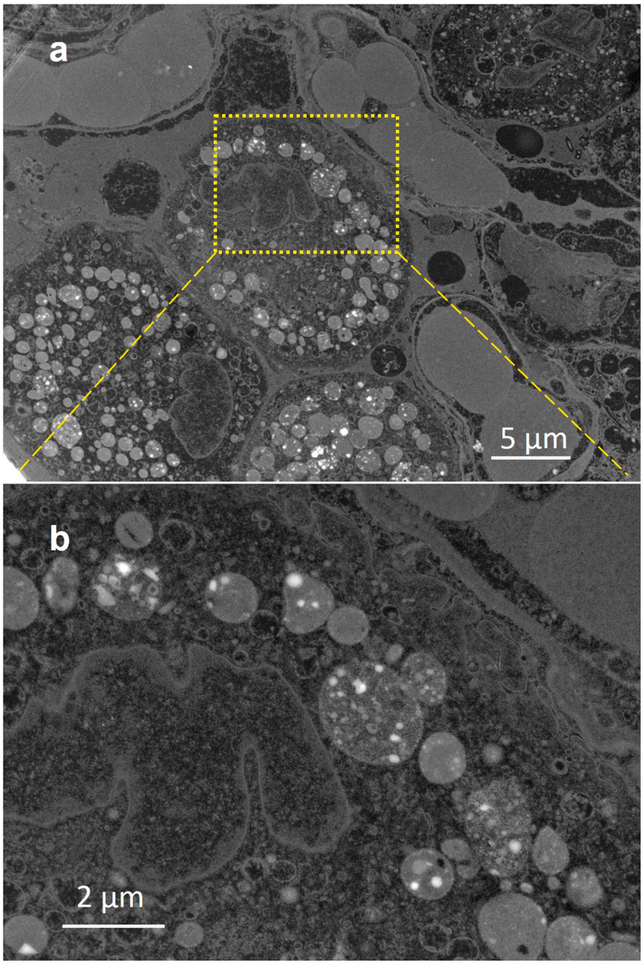

Fig. 6. Inverted images of lung cancer cells under transmission electron microscopy (TEM). (a) Numerous Fe-rich autolysosomes surround the nuclei of cancer cells that are adjacent to the rich capillary vasculature. (b) Magnified view of the highlighted region in panel (a). Ferritin particles of variable electron-density ("iron-rich" vs. iron-poor") and hemosiderin "clumps" with extreme electron-density are seen in the Fe-rich autolysosomes.

from the surface activity of dusts from iron ore mines [37,38]. Among many minerals tested, the Fe^{2+}-rich mineral berthierine was one of the most "active"—producing large amounts of free radicals within a few hours in the cell-free phosphate buffer medium. In the same oxidation tests, quartz and phyllosilicates poor in Fe^{2+} were weakly or non-active. Berthierine crystal has two basal layers: one tetrahedral layer with silicic surface and another octahedral layer with hydroxylic surface which has strong adsorption and reducing properties. The lateral surface of freshly generated berthierine also has reducing properties. The surface Fe^{2+} in berthierine may reduce molecular oxygen into activated species of oxygen such as the hydroxyl radical ($OH^{\cdot}$) or form an iron-oxo compound—for example $Fe^{IV}=O$ (ferryl, FeO^{2+})—with an equivalent oxidizing capacity [39,40]. The iron-oxo (ferryl) species in intracellular medium can attack the DNA directly [41]. Remarkably, the ferryl species can be produced $> 10^3$ times faster at boundaries between aqueous and hydrophobic media such as proteins in living tissues than the reaction in bulk water [42]. These iron species after phagocytosis could play more significant roles than hitherto envisioned in inflammatory processes underlying the development of lung cancer.

Why was this acicular Fe-rich serpentine mineral in coal missed so easily in the last four decades of lung cancer research in Xuan Wei, China? Technical difficulties in measuring these minerals in coal and coal smoke by the environmental epidemiologists in the earlier years could be one reason. Another reason could be that berthierine tends to be an overlooked mineral even by mineralogists as stated by Moore and Reynolds [18]. In the previous epidemiological research, PAHs has been a prevailing hypothesis in explaining the lung cancer epidemic of Xuan Wei since 1987 [1,43,44]; persistent free radicals [45] and gaseous carbonyl compounds [46] in coal smoke were also studied; quartz in coal smoke has also attracted a lot of attention since 2005 [3,15,17,47,48]. In reality, the detection of "many particles suggestive of welding or foundry type exposures" from lung cancer tissues of Xuan Wei [3] and the excellent correlation between Fe content in bituminous coal and lung cancer risk presented good opportunities [14], but the iron hypothesis was put aside because there was a lack of exposure-response relationship between lung cancer and the bulk concentration of iron in coal smoke according to an ecological correlation study [3]. The most recent examination of 50 lung cancer cases and 30 pseudotumor controls among non-smoking women from Xuan Wei, however, disclosed again the significant mineral deposits in the lungs and severe interstitial lung abnormalities [49]. Besides the overall iron content of minerals in coal, it might also be the acicular structure of serpentine crystal itself or some specific forms of intact or thermally modified berthierine-chamosite in coal smoke that also promote lung cancer development. Close collaboration between mineralogists and health scientists is warranted to further understand cancer risks due to household use of Fe-rich coal [50,51].

Note that local residents in the lung cancer epidemic area of Xuan Wei, China are exposed to Late Permian coal smoke, not the raw coal or coal ash characterized in this paper. Moreover, the current descriptive study was based on a limited number of human cancerous tissue samples, which should be complemented by future population-level analytical studies to establish the correspondence between mineral deposition and cancer development. As a next step, we will collect coal smoke from simulation studies in the laboratory or in the households, followed by mineralogical and toxicological analyses. The challenges include: 1) Small stove coal combustion is a very heterogeneous process which needs a lot of repeats for reproducibility; 2) The amounts of berthierine-chamosite minerals in coal smoke can be too small to meet the detection limits of semi-quantification methods such as XRD or for the purpose of chronic carcinogenesis studies. The current progress in understanding berthierine-chamosite in coal and lung cancer tissues can at least help us narrow down the study scope and fine tune our designs in coal smoke exposure assessment and carcinogenesis studies in the next steps. We ambition to finally recommend targeted environmental health policies regarding iron chelation and fuel change in afflicted areas of Xuan Wei, China, and beyond.

Environmental implication

Linkage between the Fe-rich and acicular mineral of inter-layered berthierine-chamosite in coal with lung cancer (especially among non-smoking women), uncovered in the current natural experiment setting of Xuan Wei, China, bears environmental implication beyond local intervention strategies and health protection. The study settings here could help understand the carcinogenesis mechanisms of Fe-rich fibers and contribute to evidence-based stricter regulation of all types of asbestos, including serpentine asbestos. Worldwide, the lungs are continuously exposed to Fe-containing particulate matter in the air. Our report on the iron signature of coal smoke-induced lung cancer would affect environmental health policies for air pollution management.

CRediT authorship contribution statement

Zheshen Han: Investigation, Data curation, Methodology, Writing – original draft. **Hongli Lin:** Investigation, Formal analysis. **Renyue Ji:** Data curation, Investigation, Visualization. **Lijun Liu:** Data curation, Investigation, Project administration. **Pei Zhang:** Data curation, Software. **Chao Jiang:** Software, Investigation. **Shijie Zhu:** Software, Investigation. **Jialin Xu:** Software, Investigation. **Zhenghong Yang:**

Investigation, Formal analysis. **Yunchao Huang:** Resources, Funding acquisition. **Toshihiro Kogure:** Conceptualization, Methodology, Validation. **Fang Guo:** Writing – review & editing, Validation, Project administration. **Linwei Tian:** Writing – review & editing, Conceptualization, Validation, Supervision, Resources, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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Data availability

Data will be made available on request.

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